# Technical University of Denmark

**SPRING, 2021**
**Campus: Lyngby**

**MODEL-BASED MACHINE LEARNING**

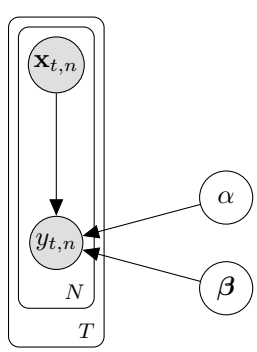
**Mini-test 2**

**(Time allowed: 90 MINUTES)**

Name: _____
Student ID: _____

## Part 1

Every year ($t = 1, \ldots, T$), a transport agency runs a survey with a fixed number of respondents, $N$. This survey includes questions on travel habits, social demographics and so on. Based on this data, it fits a binary classification model to predict whether a person bought or not a new car in that year ($y_{t,n}$) based on income and other features ($\mathbf{x}_{t,n}$). This simple model can be summarised by the following PGM and generative process:




1. Draw intercept $\alpha \sim \mathcal{N}(\alpha|0, \sigma_\alpha^2)$
2. Draw intercept $\boldsymbol{\beta} \sim \mathcal{N}(\boldsymbol{\beta}|\mathbf{0}, \sigma_\beta^2 I)$
3. For each year $t = 1, \ldots, T$
   (a) For each record $n = 1, \ldots, N$
      i. Draw class $y_{t,n} \sim \text{Bernoulli}(y_{t,n}|\text{Sigmoid}(\alpha + \boldsymbol{\beta}^T \mathbf{x}_{t,n}))$

Suppose that you discovered that the influence of the parameter $\alpha$ (the intercept of your model) changes dynamically every time step $t$, while everything else seems to be quite stable. Of course, your model needs to be adjusted accordingly - i.e., you should have a sequence of parameters $\alpha_1, \alpha_2, ..., \alpha_T$. You also found out that each $\alpha_t$ is normally distributed, and depends on the previous $\alpha_{t-1}$ and $\alpha_{t-2}$

**1.** (20%) Extend the model presented above, in order to cope with your discovery. Draw the corresponding PGM.

**2.** (10%) Write the joint distribution of your new model. Please be specific (explicitly include each distribution with parameters).

## Part 2

Consider a Gaussian Mixture Model (GMM) with 2 Gaussians with the following generative process:

1. Draw mean for cluster 1: $m_1 \sim \mathcal{N}(m_1|\mu_1, \sigma_1^2)$
2. Draw mean for cluster 2: $m_2 \sim \mathcal{N}(m_2|\mu_2, \sigma_2^2)$
3. Draw mixing proportions $\pi \sim \text{Beta}(\pi|\alpha, \beta)$
4. For each day $n \in \{1, \ldots, N\}$
   (a) Draw cluster assignment $z_n \sim \text{Bernoulli}(z_n|\pi)$
   (b) If $z_n = 1$, draw ordinary travel time $x_n \sim \mathcal{N}(x_n|m_1, 1)$
   (c) If $z_n = 0$, draw abnormal travel time $x_n \sim \mathcal{N}(x_n|m_2, 1)$

As you know, the posterior is multimodal due to the identifiability issue that is inherent to mixture models - i.e., the following two solutions:

- $\pi = 0.3$, $\mu_1 = 1$, $\sigma_1^2 = 1$, $\mu_2 = -1$, $\sigma_2 = 2$
- $\pi = 0.7$, $\mu_1 = -1$, $\sigma_1^2 = 2$, $\mu_2 = 1$, $\sigma_2 = 1$

have the exact same probability under the model.

**1.** (20%) In order to do approximate inference in this model (since exact inference is intractable), you decide to use a stochastic method. Concretely, you decided to implement the Metropolis-Hasting algorithm and to run 4 chains in parallel (to take advantage of your multi-core CPU to speed up the procedure). How would initialize the algorithm? Justify you answer.

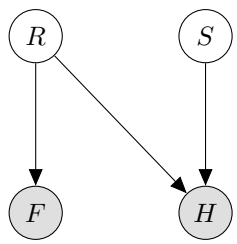
**2.** (20%) Not satisfied with the performance of Metropolis-Hasting, you decide to use a deterministic approximate inference method instead. Which type of KL divergence will you consider minimizing (i.e. $\text{KL}(q||p)$ or $\text{KL}(p||q)$)? Justify you answer.

# Part 3

One morning, when Homer leaves his house, he realises that his grass is wet ($H = 1$). He knows that it can be either due to rain ($R$), or because he has forgotten to turn off the sprinkler last night ($S$).

Next he notices that the grass of his neighbour, Flanders, is also wet ($F = 1$).

The following PGM expresses Homer's "domain knowledge" about the problem:




Let the prior probabilities for $R$ and $S$ be $P(R = 1) = 0.2$ and $P(S = 1) = 0.1$, and then the following conditional probability tables apply:

Table 1: $P(F|R)$

| | F=1 | F=0 |
|---|---|---|
| R=1 | 1 | 0 |
| R=0 | 0.2 | 0.8 |

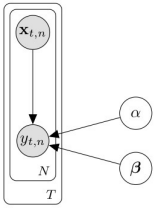
Table 2: $P(H|R, S)$

| | H=1 | H=0 |
|---|---|---|
| R=1, S=1 | 1 | 0 |
| R=0, S=1 | 0.9 | 0.1 |
| R=1, S=0 | 1 | 0 |
| R=0, S=0 | 0 | 1 |

**1.** (30%) Using Belief Propagation, calculate the probability that the grass is wet because Homer forgot to turn off the sprinkler.

# Part 1

Every year ($t = 1, \ldots, T$), a transport agency runs a survey with a fixed number of respondents, $N$. This survey includes questions on travel habits, social demographics and so on. Based on this data, it fits a binary classification model to predict whether a person bought or not a new car in that year ($y_{t,n}$) based on income and other features ($\mathbf{x}_{t,n}$). This simple model can be summarised by the following PGM and generative process:

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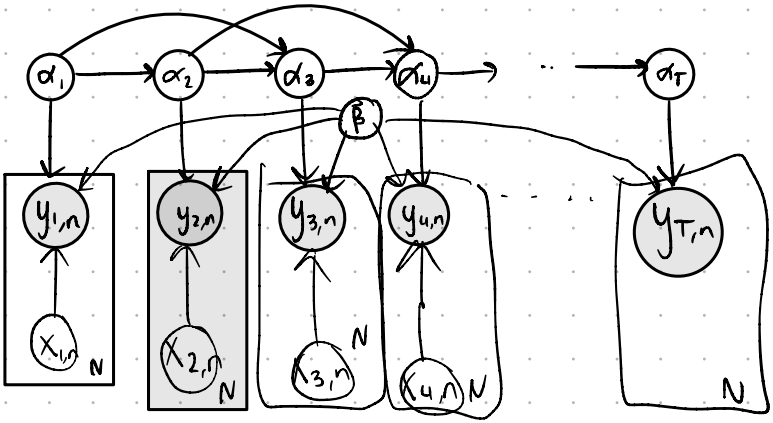


**2.** (10%) Write the joint distribution of your new model. Please be specific (explicitly include each distribution with parameters).

only $\alpha_1$ relationship

$$p(\alpha, \beta, y \mid X) = \mathcal{N}(\alpha_1 \mid 0, \sigma_\alpha^2)\, \mathcal{N}(\alpha_2 \mid \alpha_1, \sigma_\alpha^2)\, \mathcal{N}(\vec{\beta} \mid 0, \sigma_\beta^2 \vec{I}) \prod_{t=3}^{T} \mathcal{N}(\alpha_t \mid f(\alpha_{t-1}, \alpha_{t-2}), \sigma_\alpha^2) \prod_{n=1}^{N} \text{Bernoulli}(y_{t,n} \mid \text{sigmoid}(\alpha_t + \beta^T \vec{x}_{t,n}))$$

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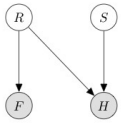


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